

Table 1 Tools for designing DNA sequence libraries and for related sequence-design problems. Rows appear in the same order as the columns of Table 2. The final two rows group tools that address problems adjacent to library design; their members are named in the table and cited individually. Tools were examined in August 2026.

Tool	Purpose	Key features	Output	Availability	Ref.
DNA Chisel	Optimize a single sequence against declared constraints and objectives	Hard constraints and scored objectives combined in one problem; codon optimization and avoidance of restriction sites, homopolymers, and extreme GC	One optimized sequence with a constraint-satisfaction report	Python package (PyPI); web application	[1]
MPRA Design Tools	Design MPRA libraries and estimate statistical power	Barcode assignment with error-correcting sets; power analysis over effect size and sequencing depth	Barcoded constructs and statistical power estimates	R package (GitHub); Shiny application	[2]
MPRAnator	Design MPRA libraries of motif arrangements and sequence variants	Combinatorial placement of motifs; randomly sampled and combinatorial substitution sets; scrambled and reverse-complement controls	Library of sequences (FASTA)	Web service	[3]
Mutation Maker	Design mutagenic primers for laboratory construction of variant libraries	Three workflows: site-scanning saturation mutagenesis, multi-site directed mutagenesis, and PCR-based gene synthesis; degenerate codons and primer mixing ratios	Mutagenic primer pairs, or gene fragments with mixing ratios	Web application (Docker); REST API	[4]
Oligopool Calculator	Design library infrastructure and analyze sequencing readout	Constraint-aware barcodes, primers, and spacers; degenerate compression of a supplied variant set; off-target screening against a host genome	Synthesis-ready library; variant counts from sequencing reads	Python package (PyPI); command line	[5]
PoolParty	Design libraries combining multiple variant types, for laboratory assays or for probing genomic models	Libraries specified as a directed acyclic graph of operations, freely mixing mutagenesis schemes; inputs from sequences, FASTA, degenerate templates, or motifs; sequences generated only on request	Library of sequences, each paired with a record of how it was constructed (CSV, TSV, FASTA, JSONL)	Python package (PyPI)	This work
tangermeme	Probe cis-regulatory logic with deep learning models	Saturation mutagenesis of input sequences; insertion and removal of motifs across sets of sequences	Perturbed sequences and model predictions	Python package (PyPI)	[6]
ValiAnT	Design and annotate libraries for saturation genome editing and cDNA deep mutational scanning	Saturation mutagenesis from genomic coordinates; several mutation types per target region; transcript-aware, including codons split across exons	Per-region libraries with metadata tables and VCF carrying variant consequence annotation	Command-line tool (source, Docker)	[7]
General-purpose toolkits (<i>Biopython</i> , <i>pydna</i> , <i>SeqPro</i>)	Manipulate sequences, simulate cloning, prepare arrays for modeling	Parsing, transformation, and file-format support	Transformed sequences	Python packages (PyPI)	[8–10]
Single-sequence design tools (<i>CodonGenie</i> , <i>ledidi</i>)	Choose codons, or edit one sequence toward a model's objective	Degenerate codons covering a target residue set; gradient-based editing guided by a predictive model	A codon set, or one edited sequence	Web service; Python package	[11, 12]

Table 2 Capabilities of tools for designing DNA sequence libraries. ● denotes a capability the tool provides; ◐ one provided with restrictions; ○ one the tool does not provide. Capabilities are counted only where the tool provides a dedicated operation, parameter, or mode. Tools were assessed in August 2026.

	DNA Chisel	MPRA Design Tools	MPRAnator	Mutation Maker	Oligopool Calculator	PoolParty	tangermeme	ValiAnT
<i>Library Specification</i>								
Library object	○	◐	○	◐	◐	●	◐	○
Composable operations	◐	○	◐	○	●	●	◐	○
Mixed variant types in one library	○	●	●	◐	○	●	○	●
<i>Variant Generation</i>								
Saturation mutagenesis	○	○	○	○	◐	●	●	●
Randomly sampled variants	●	◐	●	○	○	●	○	○
Pairwise and higher-order variants	●	○	●	●	◐	●	◐	○
Model-guided variants	●	○	○	○	○	◐	●	○
<i>Variant Types</i>								
Codon / amino-acid substitutions	●	○	○	◐	○	●	○	●
Insertions and deletions	○	●	◐	○	○	●	●	●
Combinatorial multi-motif placement	○	○	●	○	○	●	●	○
Recombination	○	○	○	○	○	●	○	○
Shuffling	○	○	◐	○	○	●	●	○
<i>Physical Construction</i>								
Synthesis-constraint checking	●	◐	◐	○	●	◐	○	◐
Constraint-based optimization	●	◐	◐	●	●	○	○	○
Primer design	◐	○	○	●	●	○	○	○
<i>Metadata and Inspection</i>								
Per-sequence design cards	●	●	◐	●	●	●	○	●
Automatic naming	○	◐	●	●	◐	●	◐	●
Sequence styling	○	○	◐	○	○	●	○	○
<i>Genomic Integration</i>								
Genome coordinates	○	◐	●	○	○	◐	◐	●
Transcript / annotation aware	○	○	○	○	○	○	○	●

References

- [1] Zulkower V, Rosser S. DNA Chisel, a versatile sequence optimizer. *Bioinformatics*. 2020;36(16):4508–4509. <https://doi.org/10.1093/bioinformatics/btaa558>.
- [2] Ghazi AR, Chen ES, Henke DM, Madan N, Edelstein LC, Shaw CA. Design tools for MPRA experiments. *Bioinformatics*. 2018;34(15):2682–2683. <https://doi.org/10.1093/bioinformatics/bty150>.
- [3] Georgakopoulos-Soares I, Jain N, Gray JM, Hemberg M. MPRAinator: a web-based tool for the design of massively parallel reporter assay experiments. *Bioinformatics*. 2017;33(1):137–138. <https://doi.org/10.1093/bioinformatics/btw584>.
- [4] Hiraga K, Mejzlik P, Marcisin M, Vostrosablin N, Gromek A, Arnold J, et al. Mutation Maker, An Open Source Oligo Design Platform for Protein Engineering. *ACS Synth Biol*. 2021;10(2):357–370. <https://doi.org/10.1021/acssynbio.0c00542>.
- [5] Hossain A, Cetnar DP, LaFleur TL, McLellan JR, Salis HM. Automated Design of Oligopools and Rapid Analysis of Massively Parallel Barcoded Measurements. *ACS Synth Biol*. 2024;13(12):4218–4232. <https://doi.org/10.1021/acssynbio.4c00661>.
- [6] Schreiber J. tangermeme: A toolkit for understanding cis-regulatory logic using deep learning models. *bioRxiv*. 2025;p. 2025.08.08.669296. <https://doi.org/10.1101/2025.08.08.669296>.
- [7] Barbon L, Offord V, Radford EJ, Butler AP, Gerety SS, Adams DJ, et al. Variant Library Annotation Tool (VaLiAnT): an oligonucleotide library design and annotation tool for saturation genome editing and other deep mutational scanning experiments. *Bioinformatics*. 2022;38(4):892–899. <https://doi.org/10.1093/>

[bioinformatics/btab776](#).

- [8] Cock PJA, Antao T, Chang JT, Chapman BA, Cox CJ, Dalke A, et al. Biopython: freely available Python tools for computational molecular biology and bioinformatics. *Bioinformatics*. 2009;25(11):1422–1423. <https://doi.org/10.1093/bioinformatics/btp163>.
- [9] Pereira F, Azevedo F, Carvalho Â, Ribeiro GF, Budde MW, Johansson B. Pydna: a simulation and documentation tool for DNA assembly strategies using python. *BMC Bioinformatics*. 2015;16(1):142. <https://doi.org/10.1186/s12859-015-0544-x>.
- [10] Klie A, Laub D. SeqPro (Sequence processing toolkit); 2023. Software. Available from: <https://github.com/ML4GLand/SeqPro>.
- [11] Swainston N, Currin A, Green L, Breitling R, Day PJ, Kell DB. CodonGenie: optimised ambiguous codon design tools. *PeerJ Comput Sci*. 2017;3:e120. <https://doi.org/10.7717/peerj-cs.120>.
- [12] Schreiber J, Lorbeer FK, Heinzl M, Lu YY, Stark A, Noble WS. Programmatic design and editing of cis-regulatory elements. *bioRxiv*. 2025;p. 2025.04.22.650035. <https://doi.org/10.1101/2025.04.22.650035>.